EXPERIMENT-3:

OUTPUT:

First Five Records

	PassengerId	Survived	Pclass	...	Fare	Cabin	Embarked
0	1	0	3	...	7.2500	NaN	S
1	2	1	1	...	71.2833	C85	C
2	3	1	3	...	7.9250	NaN	S
3	4	1	1	...	53.1000	C123	S
4	5	0	3	...	8.0500	NaN	S

[5 rows x 12 columns]

Dataset Shape: (891, 12)

Data Types

PassengerId	int64
Survived	int64
Pclass	int64
Name	str
Sex	str
Age	float64
SibSp	int64
Parch	int64
Ticket	str
Fare	float64
Cabin	str
Embarked	str

dtype: object

Missing Values

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Missing Values After Imputation
PassengerId   0
Survived   0
Pclass   0
Name   0
Sex   0
Age   0
SibSp   0
Parch   0
Ticket   0
Fare   0
Cabin   0
Embarked   0
dtype: int64

Duplicate Records: 0
Dataset Shape After Removing Duplicates: (891, 12)

Encoded Data
   Sex Embarked
0   1   2
1   0   0
2   0   2
3   0   2
4   1   2

Standardized Age and Fare
        Age   Fare
0 -0.592481 -0.502445
1 0.638789 0.786845
2 -0.284663 -0.488854
3 0.407926 0.420730
4 0.407926 -0.486337

Normalized Age and Fare
        Age   Fare
0 0.271174 0.014151
1 0.472229 0.139136
2 0.321438 0.015469
3 0.434531 0.103644
4 0.434531 0.015713

Number of Outliers in Fare: 116
Dataset Shape After Removing Outliers
(775, 12)

Final Dataset

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	1	0.271174	1	0	0.014151	2
2	1	3	0	0.321438	0	0	0.015469	2
3	1	1	0	0.434531	1	0	0.103644	2
4	0	3	1	0.434531	0	0	0.015713	2
5	0	3	1	0.367921	0	0	0.016510	1